

Troubleshoot the Map Function

NOTE

Aggregation Pipeline as Alternative to Map-Reduce

Starting in MongoDB 5.0, map-reduce is deprecated:

- Instead of map-reduce, you should use an aggregation pipeline. Aggregation pipelines provide better performance and usability than map-reduce.
- You can rewrite map-reduce operations using aggregation pipeline stages, such as `$group`, `$merge`, and others.
- For map-reduce operations that require custom functionality, you can use the `$accumulator` and `$function` aggregation operators. You can use those operators to define custom aggregation expressions in JavaScript.

For examples of aggregation pipeline alternatives to map-reduce, see:

- [Map-Reduce to Aggregation Pipeline](#)
- [Map-Reduce Examples](#)

An aggregation pipeline is also easier to troubleshoot than a map-reduce operation.

The `map` function is a JavaScript function that associates or "maps" a value with a key and emits the key and value pair during a map-reduce operation.

Verify Key and Value Pairs

To verify the `key` and `value` pairs emitted by the `map` function, write your own `emit` function.

Consider a collection `orders` that contains documents of the following prototype:

```
{
  _id: ObjectId("50a8240b927d5d8b5891743c"),
  cust_id: "abc123",
  ord_date: new Date("Oct 04, 2012"),
  status: 'A',
  price: 250,
  items: [ { sku: "mmm", qty: 5, price: 2.5 },
    { sku: "nnn", qty: 5, price: 2.5 } ]
}
```

1. Define the `map` function that maps the `price` to the `cust_id` for each document and emits the `cust_id` and `price` pair:

```
var map = function() {
  emit(this.cust_id, this.price);
}
```

```
};
```

2. Define the `emit` function to print the key and value:

```
var emit = function(key, value) {  
  print("emit");  
  print("key: " + key + " value: " + toJson(value));  
}
```

3. Invoke the `map` function with a single document from the `orders` collection:

```
var myDoc = db.orders.findOne( { _id:  
  ObjectId("50a8240b927d5d8b5891743c") } );  
map.apply(myDoc);
```

4. Verify the key and value pair is as you expected.

```
emit  
key: abc123 value:250
```

5. Invoke the `map` function with multiple documents from the `orders` collection:

```
var myCursor = db.orders.find( { cust_id: "abc123" } );  
while (myCursor.hasNext()) {  
  var doc = myCursor.next();  
  print ("document _id= " + toJson(doc._id));  
  map.apply(doc);  
  print();  
}
```

6. Verify the key and value pairs are as you expected.

TIP

See also:

The `map` function must meet various requirements. For a list of all the requirements for the `map` function, see `mapReduce`, or `mongosh` helper method `db.collection.mapReduce()`.